

# Leveraging Animal Behavior in a Robot Plan\*

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**Abstract**—Poultry farming is labor-intensive, requiring farmers to perform daily tasks such as removing deceased birds, collecting floor eggs, and assessing chickens. These tasks contribute to high turnover rates in the industry, averaging 65% and reaching up to 150% in some locations. To address this, we developed an autonomous robot capable of performing daily walkthroughs of the farm. Our goal is to enable robots to navigate densely packed chicken houses without harming or unnecessarily stressing the birds. We tested two navigation approaches: one that avoids the chickens as static obstacles and another that nudges them to move. In commercial broiler-breeder farms, the avoid-only method proved ineffective due to frequent interruptions by chickens, while the interactive method achieved a 100% success rate. Increasing the nudging speed from 0.3 m/s to 0.45 m/s improved the traversal distance by 47.8% in the morning and 37.1% in the afternoon. These findings suggest that interactive navigation significantly enhances robot performance in navigating crowded spaces.

## I. INTRODUCTION

Robots operating within dense populations of animals in confined spaces is a challenging and largely unexplored research area. When considering the breadth of research with robots and animals, it is mostly confined to controlling flocks and herds [1], [2], [3], performing stationary tasks such as automated milking [4], mimicking animals with bio-inspired designs [5], [6], and passively observing the animals [7]. However, these researches do not focus on the robots' ability to navigate the environment, nor do they address operations in confined spaces.

According to the USDA agricultural survey conducted in 2017, there were 164,099 poultry farms in the United States that produced over 8.9 billion chickens [8] for a total of 49.2 billion in sales. On these farms, farmers and farm hands must walk through the flocks daily, performing tasks such as removing mortality, picking up floor eggs, and qualitatively assessing the birds. These tasks are labor intensive and result in high turnover rate.

In 2018, The U.S. Poultry and Egg Association surveyed 40 different locations in the industry [9] and found that the average turnover rate was 65% with some locations reporting turnover rate as high as 150%. This rate of turnover is excessive when compared to the average turnover rate in the industry, which is 30.4%. It is clear from these statistics that the industry would benefit from automation and technologies to help reduce the labor requirements. Robotics is one such possibility for enabling automation within the poultry farm.

For robots to operate in these types of environments, they must first be able to successfully navigate them without negatively affecting the welfare of the animals. Over the last 20 years, an increasing number of people are demanding ethical production systems and claim to refuse to buy products that do not meet their animal welfare standard [10]. Simply stated, the robot absolutely must not harm the animals to maintain positive public opinion and support a sustainable ethical production system.



Fig. 1. Our mobile robot operating in a densely packed commercial broiler-breeder chicken farm with 9600 chickens confined in a 120 m by 15 m area.

The goal of this work is to enable a robot to successfully navigate through a densely packed commercial chicken house without harming or unnecessarily stressing the birds. An example of the densely packed commercial chicken house can be seen in Fig. 1. In this paper, we make the following contributions: (i) We identify a robot interaction that elicits a repeatable response from the animal. (ii) We present a novel method leveraging animal behavior as part of the robot plan via this interaction. (iii) We contrast the interaction method with a simple avoid-only method and evaluate it in a real-world setting.

In this research, two methods of navigation are implemented and tested: simple navigation via avoid-only approach, which avoid any interaction with the chickens, and a method where we interact with the birds by nudging them, causing them to consistently move away from the robot. Here we again stress the importance of navigating the flock without negatively affecting the welfare of the animal.

## II. BACKGROUND

### A. Navigation Through Crowded Environment

Mobile robots navigating through crowded environment has been studied in the context of dense human crowd. Traditional approach to the problem treats human agents as dynamic obstacles during path planning and navigation [11],

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[12]. However, this approach encounters the "freezing robot" problem when the crowd density surpass a certain threshold. In these cases, the high density of the crowd renders all path infeasible. Various improvement to this approach have modeled and predicted pedestrian trajectory to improve the success rate of the navigation [13], [14], [15]. Without directly interacting with agents in the crowd, this navigation approach still fails when density of the crowd are very high.

More recently, machine learning approaches have been leveraged to solve this navigation problem. Deep reinforcement learning has been applied extensively to enable mobile robots to navigate in crowded environment [16], [17], [18], [19]. But, the focus of these approaches is often placed on minimizing the interaction with the human agents to avoid potential conflicts. In many cases, when the density of crowd surpass a certain threshold, it is difficult or impossible to reach the goal without interacting with people.

### B. Navigation While Leverage Interaction

Leveraging verbal and nonverbal interaction with human agents allows mobile robots to efficiently and safely navigate complex environments [20], [21], [22]. Combining verbal communications with social-aware navigation can improve the success rate and safety of navigation in crowded environments [23]. Nonverbal gestures can communicate robot intent to facilitate navigation around human agents [24]. Dugas et al. investigated a high level planning algorithm that leverages different interaction modalities, such as saying and nudging, for navigating in dynamic and crowded environment [25]. These interactions are strictly limited to human-robot interaction and have limited applicability to robot-animal interactions due to differences in human and animal behaviors.

### C. Poultry Robots

The impact on chicken behavior is minimal when mobile robot are operated safely [26]. Limited disruption to chickens' behavior was found in a preliminary study on chicken's behavioral response to mobile robot [27]. Increasing chickens' exposure to robot and reducing robot's movement speed lowered the distance at which the chickens start avoiding the approaching object [28]. A study found that chicken can become more sedentary after the feeding period [29]. This can lead to chickens not actively avoiding the robot during its navigation. To minimize robot-induced stress and ensure the safety of chickens, the behavior of the chickens should be taken into consideration during the design and deployment of mobile robots in the chicken farm.

Various robot platforms have been developed to operate safely in poultry farms [30], [31], [32]. Lei et al. developed and evaluated a path planning algorithm in simulation to search for and retrieve deceased chickens in the farm [33]. Vroegindewij et al. treated chickens as static obstacles when evaluating their egg collection robot platform inside of a commercial chicken farm [34]. The robot platform required manual interventions to correct errors, such as collisions. Autonomous safe navigation through crowded animal spaces

was not explored in these mobile robots. We developed a novel path planning and navigation framework that leverages animal behavior as part of its navigation algorithm.

## III. MATERIALS AND METHODS

### A. Robot Platform

For this effort, a mobile robot platform was developed using a combination of custom and commercial-off-the-shelf parts. The robot chassis is a TP-140-042 from Super Droid Robots, consisting of an aluminum body and four independent brushed DC motors (IG-42 078 rpm gear motor with encoder). The robot is custom fitted with Trampa Tractor Tread 10-inch tires with a wider contact area for enhanced grip which is necessary for operation on wood-chipped floors common in poultry houses. A bumper and various sensor mounts are fabricated out of 3D-printed parts. An image of the robot is shown in Figure 2.

Two independent Roboteq SDC2160 2-channel brushed DC motors are used, one controlling the front left and right wheels, and the second controlling the rear left and right. The IG-42 motor encoders are integrated via these controllers. The robot is powered by a 16-amp hour lithium polymer battery connected to a Roboteq BMS1040x battery management system.

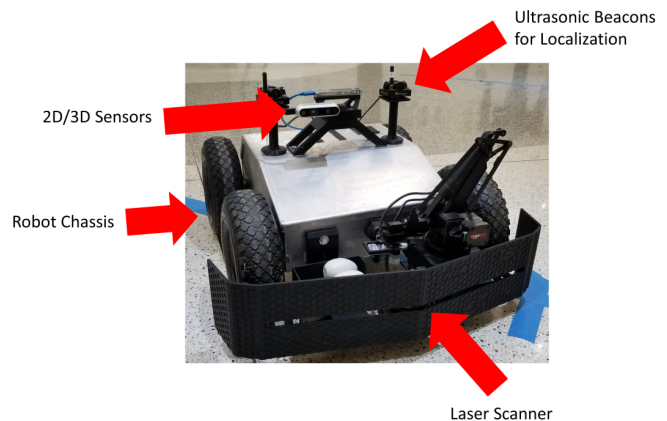


Fig. 2. Physical robot platform.

The sensor payload on the robot consists of an Intel RealSense D435 stereo camera, an RPLidar A2 360-degree 2D laser scanner, a Phidgets 3/3/3 inertial measurement unit and two ultrasonic-based localization beacons from Marvelmind. The RealSense camera is mounted on top of the robot while the laser scanner is mounted inside the front bumper. A slit in the front bumper allows for a window for the laser scanner to see out. For this study, the RealSense camera data is not utilized and only the laser scanner is used. The ultrasonic localization beacons work alongside beacons installed in the infrastructure to enable localization accuracy of approximately  $\pm 2$  inches in the chicken house. The beacons are installed on top of the robot chassis approximately 12 inches apart. Using two beacons, the system can resolve both position and heading simultaneously during operation.

All the electronics are connected via USB to an Nvidia AGX Xavier developer kit which serves as the robot controller.

### B. Robot Operation

The robot control utilizes the Robot Operating System [35] as a basic framework with both open-source and custom packages. The encoders, IMU, and localization beacons are all fused via an extended kalman filter using the Robot Pose EKF package. Both ROS Navigation Stack [36] and the actionlib provide the foundation for the control logic which provides path planning and activity of the robot during operation. An occupancy grid map via the ROS Navigation Stack is used to enable localization and path planning. The physical location of the sensors and the robot dimensions are encoded in a URDF file and the ROS tf2 transform library is used to manage all of the various transforms required for operation.

Global path planning is performed in the global map and local planning and interactions are performed in the local map. The global path planning uses Dijkstra's algorithm. Dynamically detected obstacles are placed into the local map, allowing the robot to modify local plans without having to re-plan the larger route to the goal.

The map is customized based on the chicken house layout, where the floor area is typically a simple rectangular shape. The map is represented with an image, where each pixel represents 5 square centimeters. Each pixel in the map image represents navigable area and static obstacle locations which in the case of this study only consist of the 4 walls enclosing the space.

Chickens and static obstacles are detected using the bumper-mounted laser scanner. An open-source ROS package for 2D obstacle detection based on laser range data is used to group the laser scans into discrete obstacles. A custom ROS node subscribes to the obstacle detector messages which provides a list of detected obstacles. The closest obstacle is determined by calculating the distance of all detected obstacles relative to the front of the robot and finding the minimum. A lookup into the occupancy map then checks if the closest obstacle is part of the static obstacles or not. If it is not part of the static obstacles, it is assumed to be a chicken.

### C. Navigation Algorithms

Successful navigation in this context means the ability for the robot to reach its desired waypoints while navigating through the flocks of birds without harming them. Therefore, two separate navigation approaches were developed attempting to leverage the behavior of the chickens. As one advances through the house, the chickens tend to move out of the way. However, as chickens get more comfortable or are otherwise inattentive, they will not necessarily move out of the way. This leads to our hypothesis that navigation via avoiding only will be unsuccessful in the house, and the robot will be required to safely interact with the birds in order to achieve its goals. This leads to the following navigation approaches presented below.

**Navigation by Avoiding Only:** Under this method, the robot is programmed to attempt to find a path to its goal while avoiding any contact or interaction with the chickens under any circumstance. Essentially, the robot searches for an available path where there are no chickens blocking the way and advances along any identified route. If it is blocked by chickens, it will continuously search around itself for a new opening to exploit until it is able to reach the goal. A state diagram of this is presented in Fig. 3.

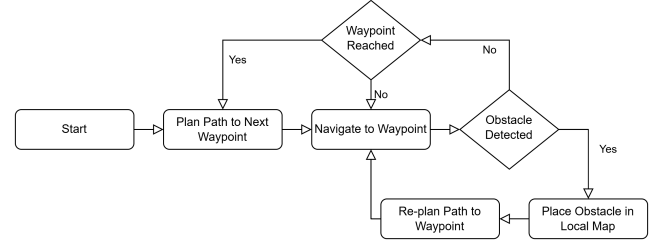


Fig. 3. State diagram for navigation by avoid only method.

**Navigation with Interaction:** As an alternative to navigation by avoiding only, we propose a method for navigation where we interact with the animals while leveraging their behavior in order to improve the chances of success. In this case, the interaction is simply nudging the chickens with the bumper of the robot to encourage them to move. When an obstacle (chicken) is detected, the robot will stop and wait for a period of two seconds to allow the chicken time to move away from the robot. If after this time, the chicken is still blocking the robot path, the robot will nudge the chicken to encourage it to move. This is achieved by calculating the distance between the robot and the chicken and initiating a custom routine where the robot advances until it contacts the chicken, gently nudging it, and then backs up to its original pre-nudging position to scan the environment and see if the chicken has moved. If the chicken has moved, the robot then advances. If the chicken has not moved, the robot will attempt a second nudge. Failing to move the chicken after two attempts will result in the robot attempting to circumnavigate via placing an obstacle in the map and re-planning a route around the chicken. The state diagram for this behavior is shown in Fig. 4.

### D. Test Facilities

The test facility is a commercial broiler-breeder chicken farm with 9600 chickens of breeding age (22-24 weeks) consisting of 8500 females and 900 males. It is approximately 120 meters long and 15 meters wide, or 1800 m<sup>2</sup>. Following a conventional layout for these houses, the house contains nests on raised slats on either side of the house and central scratch area that allows for free movement of the chickens. The robot operates in this scratch area which is approximately 4.55 m wide and 114 m long. An image of the test farm is shown in Fig. 5.

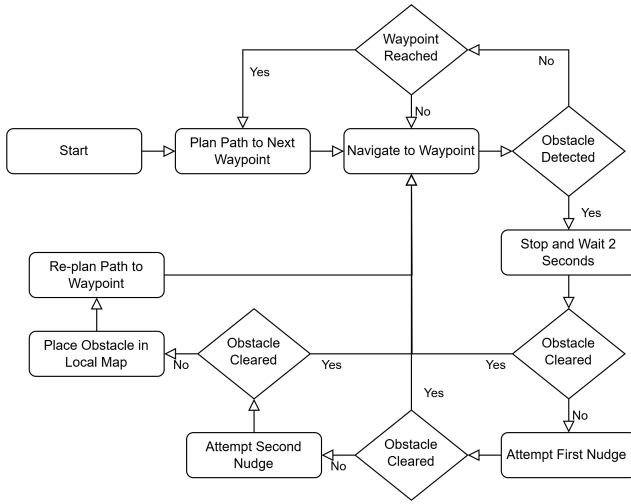


Fig. 4. State diagram for navigation with nudging behavior.



Fig. 5. Image of Test Farm.

#### IV. EXPERIMENTAL PLAN

For each of the navigation methods (nudge vs. avoid), the robot will start on one end of the scratch area in the chicken house and will be given a goal waypoint at the opposite end of the house. During operation, the time, location, and action type of the robot is recorded for each action that the robot performs. Actions include: 1) stopping due to obstacle detected 2) nudging obstacle 3) continuing movement due to obstacle cleared 4) circumnavigating due to failed nudging. For each action, the location of all detected obstacles within 1 meter in front of the robot is also recorded. In addition, the location of the robot in the house is recorded at a rate of 1 Hz as is the status of the battery system including voltage and instantaneous current draw at a rate of approximately 1.5 Hz. Three specific studies were carried out under this effort.

##### A. Animal Reaction as a Function of Nudging Speed

The goal of this study is to determine if there exists any statistically significant difference in the behavior of the chickens based on the aggressiveness of the nudging operation performed by the robot. This is achieved by gradually increasing the velocity of the robot during the custom nudge behavior. The higher the speed, the higher the aggressiveness of the nudge will be, as the robot will impact the chicken with a greater force. Three levels of velocity are tested: 0.3,

0.45, and 0.6 m/s. The robot performs 30 nudges under each velocity setting for evaluation.

##### B. Comparison of Navigation Method's Effectiveness

The robot is observed operating for a period of 10 minutes while attempting to navigate. This test will be repeated 10 times for each method to attempt to characterize repeatability given the random nature of the animals. The total distance traversed, the number and type of interactions, and the cost of transport will be evaluated.

##### C. End-to-End Navigation of the Entire Facility

Traversal of the entire house from end-to-end will be attempted to determine the efficacy of each navigation method. As before, the number and type of interactions will be recorded for each attempt. In addition, the total time required to traverse (if successful), the energy consumed, and the cost of transport will be evaluated.

#### V. RESULTS AND DISCUSSION

Our tests showed that the behavior of chickens was significantly different at different times of the day. A literature review [29] confirms that bird activity changes throughout the day. Early in the day, chickens are more active, mobile and energetic before and shortly after feeding. They become more sedentary and prefer to sit down later in the day. This was clear in our tests and is also reflected in our data. Therefore, we also attempt to separate the results from the navigation with respect to the time-of-day and gross behavior change of the chickens. Fig. 6 illustrates this change in behavior and density showing both images of the robot and the relative densities of the chickens before and after nudges.

##### A. Animal Reaction as a Function of Nudging Speed

Two 30-nudge tests at each speed were executed, once in the morning and once in the afternoon, to account for the observed variations in bird behavior mentioned above. While the robot's ability to traverse the house showed significant differences between morning and afternoon runs, the relative change in bird behavior with respect to nudging speed remained fairly uniform across both time periods. Bird displacement, as defined here, refers to the distance a bird or group of birds moves following an interaction with the robot. When considering the displacement of the closest bird after being nudged, a one-way ANOVA (Fig. 7) on the morning test shows significant differences in bird displacement at different robot nudging speeds. On average, the displacement of the closest chicken after being nudged is higher when they are nudged at a speed of 0.45 m/s ( $p = 0.0001$ ) or 0.60 m/s ( $p < 0.0001$ ) compared to 0.30 m/s. No significant difference was observed when the bird was nudged at a speed between 0.45 and 0.60 m/s ( $p = 0.9664$ ).

The afternoon results mirror those of the morning, with a significant difference in the displacement of the closest chicken, which is higher when nudged at a speed of 0.45 m/s ( $p = 0.0274$ ) and 0.60 m/s ( $p = 0.0088$ ) compared to 0.3 m/s on average. Likewise, no significant difference is observed between the 0.45 m/s and 0.6 m/s nudges ( $p = 0.9460$ ).



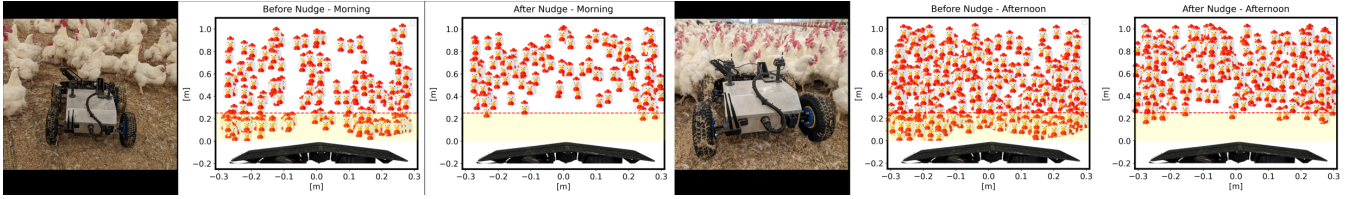


Fig. 6. From left to right: 1) Image of robot in morning with active and less dense birds in scratch area 2) Cumulative plot of location of birds relative to front of robot before nudging in the morning 3) Cumulative plot of location of birds relative to front of robot after nudging in the morning 4) Image of robot in afternoon with less active and dense birds in scratch area 5) Cumulative plot of location of birds relative to front of robot before nudging in the afternoon 6) Cumulative plot of location of birds relative to front of robot after nudging in the afternoon.

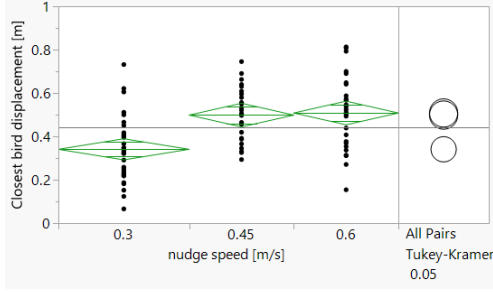


Fig. 7. One-way ANOVA analysis for displacement of closest bird to robot after nudging in the morning.

Although the one-way ANOVA does not indicate a significant difference in displacement of closet bird between 0.45 m/s and 0.60 m/s nudge speed, results on the distance traveled by the robot provided another perspective on the navigation performance. Across the six tests, it is clear that as the nudge velocity increases, the robot is able to traverse a greater distance in the house. Fig. 8 shows the distances traversed for each test. In the morning, the robot travelled 20.9, 30.9, and 44.7 meters with nudge speeds of 3.0, 4.5, and 6.0 m/s respectively. In the afternoon, the robot travelled 6.2, 8.5, and 13.4 meters. This resulted in a gain of 47.8% from 0.45 vs. 0.3 in the morning, and 37.1% gain in the afternoon. Similarly, the gain from 0.60 vs. 0.45 was 44.7% in the morning and 57.6% in the afternoon. This indicates that the relative changes in behavior due to different nudge speeds are similar.

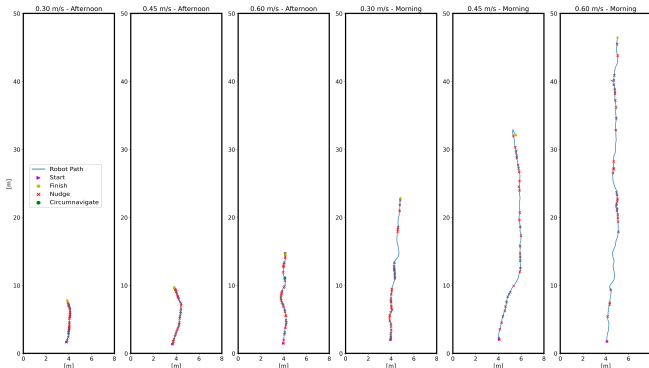


Fig. 8. Distance traversed in house by robot as a function of nudge speed and time of day.

We hypothesize that the reason the robot is able to move farther, even though the average displacements of the closest birds are not statistically different, is that the more aggressive the nudging behavior, the more it affects all of the birds in a larger area in front of the robot. Fig. 9 shows the results of a one-way ANOVA on the average displacement of all birds from the robot after a nudge, not just the closest bird.

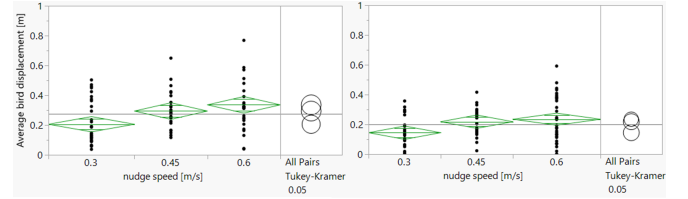


Fig. 9. One way ANOVA for average displacement of all birds in front of robot after a nudge for morning (left) and afternoon (right).

We did not see a significant difference between nudges with speeds of 0.45 m/s and 0.6 m/s in the morning ( $p = 0.5428$ ) or the afternoon ( $p = 0.8471$ ). There is also no significant difference between the displacement of all birds in front of the robot between 0.3 and 0.45 in the morning ( $p = 0.0581$ ) or the afternoon ( $p = 0.0588$ ). Looking further, Table I provides additional insight, showing a general reduction in the average number of nudges per meter with each increase in nudge speed. This indicates that the more aggressive nudge results in a larger affected area in front of the robot, causing more chicken in a larger area to move out of the robot path. This behavior is reinforced with qualitative observation during each of the tests, where it was clear that more aggressive nudging resulted in a larger affected area in front of the robot.

TABLE I  
NUMBER OF CHICKENS IN FRONT OF ROBOT AFTER A NUDGE

| Nudge Speed<br>(m/s) | Number of Birds |          | Nudges/m<br>(m) |
|----------------------|-----------------|----------|-----------------|
| 0.3                  | $\mu$           | $\sigma$ |                 |
| 0.3                  | 2.0             | 0.88     | 1.72            |
| 0.45                 | 1.5             | 0.87     | 1.1             |
| 0.6                  | 1.6             | 0.93     | 0.74            |

While these tests indicate that a nudging velocity of 0.60 m/s results in the largest clearing of obstacles and the most distance traversed, allowing for the more efficient movement, it was observed that some chickens squawked audibly and

flapped wings when the robot impacted them. This behavior indicates an interaction that may cause stress to the bird [37] and was therefore deemed unacceptable for our purposes. A nudging velocity of 0.45 m/s is chosen as it both optimizes the efficiency of the robot while having no indicators of additional stress induced on the animals.

### B. Comparison of Navigation Methods Effectiveness

For the avoid-only method, the robot was constantly interrupted by chickens blocking the path regardless of the time of day. While there was more success in traversing the house in the morning, the robot still was unable to make any significant progress. When surrounded by chickens, the robot would perform an in-place turn, attempting to turn away from the chickens to find an alternative route. This resulted in the robot repeatedly circling 360-degrees as it searched for an opening to navigate.

For each of the 10 tests performed, it was clearly shown that the navigation via avoid-only method was unsuccessful. While it was able to navigate 18 meters in one of the tests, it did not make it past 2 meters in 5 of the tests. Fig. 10 shows the results from the longest traversal of the robot with the avoid-only navigation method.

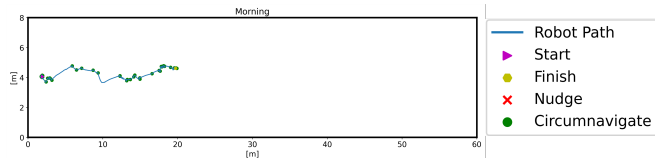


Fig. 10. Sample plot from longest distance navigated by robot with avoid only method

Navigation with the nudge interaction performed better. This interaction allowed the robot to robustly and consistently advance through the house. Across all tests, the robot was able to navigate 41.2 meters on average ( $\sigma = 10.8$ ). Considering just the tests in the morning, the average increases to 49.8 meters ( $\sigma = 8.05$ ), with 34.1 meters ( $\sigma = 8.5$ ) on average in the afternoon. For all tests, the robot was never stuck and continued successful navigation until the test was terminated at the 10 minutes mark for each replication. Fig. 11 shows the results from the longest traversal of the robot with the navigation with interaction method.

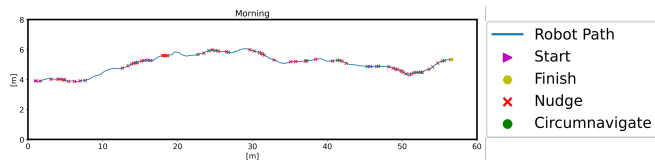


Fig. 11. Sample plot from longest distance navigated by robot with navigation with interaction

### C. End-to-End Navigation of the Entire Facility

Attempts to navigate the entire facility for the navigation via avoid-only method was attempted once and was unsuccessful with the robot only navigated 5.7 meters of the floor.

In addition, results from the 10-minutes run tests showed the robot failing to advance any significant amount in the house using this method and we deemed it unnecessary to make further attempts to navigate the entire house. For the navigation with interaction method, 3 attempts were made with the robot successfully reaching the goal 109 meters away in an average time of 21.6 minutes ( $\sigma 4.5$ ) for a 100% success rate. The path taken by the robot with location and frequency of types of interactions is shown in Fig. 12.

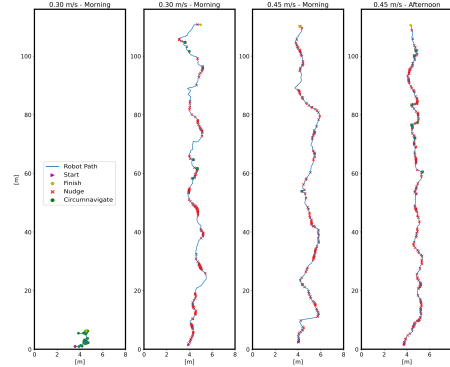


Fig. 12. Distance traversed by robot for whole house navigation attempts

Table II summarizes the average number of interactions by type that the robot encountered during navigation over all three attempts. In all attempts, the robot was able to successfully reach the goal. Over the three tests, the robot was required to nudge the birds on average 1.55 times per meter of distance navigated.

TABLE II  
INTERACTION FREQUENCY BY TYPE DURING WHOLE-HOUSE NAVIGATION

| Interaction Type | $\mu$ | $\sigma$ |
|------------------|-------|----------|
| Stop/Wait        | 291   | 41.3     |
| Nudge            | 169   | 27.1     |
| Circumnavigate   | 9.7   | 8.5      |
| Advance          | 281   | 32.3     |

## VI. CONCLUSIONS

We presented in this paper a method enabling safe navigation in a densely packed commercial poultry house. Safe navigation in this case means the ability for a robot to reach a goal without harming or negatively affecting the welfare of the animals in its environment. In doing so, we compared and contrasted two methods of attempting to achieve this goal: (1) Navigation by avoiding any interaction with the animals, and (2) Navigation while interacting (nudging) the animals to encourage them to move out of the way of the robot. We showed that the robot system was unable to reach the goal state when only allowed to avoid the animals, but was able to successfully reach the goal state 100% of the time when programmed to nudge the animals. We also successfully quantify the increase in robots' navigation performance when the nudge speed was changed between 0.30, 0.45, and 0.60 m/s.

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